

# A Universal Geometric Theory of Nuclear Reactions in CPTG

The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation

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## Abstract

Primordial nucleosynthesis is normally represented as a high-dimensional coupled reaction network in which individual reaction rates are propagated numerically to final abundances. This paper develops a lower-dimensional geometric formulation within Curvature Polarization Transport Gravity (CPTG). The light-nuclide backbone is organized into four ordered sectors—vertex, bridge, closure, and saturation:

$$(n, p) \rightarrow D \rightarrow (T, {}^3\text{He}) \rightarrow {}^4\text{He}.$$

The construction separates a fixed scalar curvature-response law, an exact baryon-conserving reaction-source geometry determined by nuclear stoichiometry, and a network-native dynamical pushforward to observable abundances. Three ordered transport directions and one charge-constrained polarization direction form an exact rank-four reaction-source coordinate in the declared light-sector domain. Reaction directions are therefore fixed geometrically before numerical endpoint responses are examined, rather than inferred from them. The nuclear networks remain the dynamical calculators: CPTG fixes the source geometry, while each implementation retains its own rates, thermodynamics, Jacobian, and abundance evolution.

PRIMAT 0.3.2 provides the primary numerical test of the first-order source-to-endpoint map: all 48/48 primary resolved source/Jacobian predictions pass, with worst endpoint-vector discrepancy 0.0367629% and minimum direction cosine 0.99999999209394. In the shared Reaction-20 logarithmic-response coordinate ( $D/H$ ,  ${}^3\text{He}/H$ ,  ${}^7\text{Li}/H$ ), the median native vectors are  $(+0.343898, -0.387450, -0.602684)$  for PRIMAT,  $(+0.342815, -0.386472, -0.608490)$  for PArthENoPE, and  $(+0.343241, -0.387065, -0.601891)$  for PRyMordial. Across the three networks, the median directions agree at cosines exceeding 0.99998. The independently fixed mass-seven transport law passes all 23/23 preregistered PRIMAT gates, with survival differing from  $1/\pi$  by 0.0177852%. Together, these results support a compact geometric source-to-observable architecture that preserves native-network dynamics without reaction-specific geometric fitting or a universal code-independent susceptibility matrix.

# 1 Introduction

Primordial nucleosynthesis is normally described as a coupled network of nuclear reactions: individual thermonuclear rates act on an evolving abundance vector, and the final light-nuclide yields emerge only after the entire system has been propagated through the thermal history of the early Universe [7, 8, 9, 10]. That framework is indispensable for precision calculation, but it leaves a separate structural question open. Can the response of many different reactions be organized by a single geometric law, rather than assigning an unrelated effective rule to each reaction channel? This paper develops and tests such a formulation within Curvature Polarization Transport Gravity (CPTG) [1].

The central construction is a universal four-sector geometry for the primordial light-nuclide backbone. The evolving nuclear state is coarse-grained into the ordered sequence

$$(n, p) \rightarrow \text{D} \rightarrow (T, {}^3\text{He}) \rightarrow {}^4\text{He}, \quad (1)$$

with the corresponding structural interpretation

$$\text{vertex} \rightarrow \text{bridge} \rightarrow \text{closure} \rightarrow \text{saturation}. \quad (2)$$

The free proton–neutron pair forms the vertex sector, deuterium is the first bonded bridge, the tritium–helium-three mirror pair forms the mass-three closure sector, and helium-four is the saturated terminal state of this four-sector light-element domain. These sectors are not intended to replace the underlying reaction network or to imply a single literal reaction chain. They define a common state geometry through which the reactions of the native network can be represented.

What distinguishes the theory is the separation of three objects that are usually entangled in a numerical abundance calculation. First, CPTG supplies a fixed scalar curvature-response law that sets the local response scale. Second, exact nuclear stoichiometry determines the direction in which each physical reaction injects transport and charge polarization into the four-sector state. Third, the native reaction network determines how that source perturbation propagates to measurable endpoint abundances. The geometric law is therefore not refitted from reaction to reaction, while the physical identity of each reaction is retained exactly through its stoichiometric source direction and the full network retains its own dynamical response.

This separation is important because it makes the theory predictive in a stricter sense than an abundance fit. Once the scalar law and source geometry are fixed, a reaction perturbation determines a source vector before the corresponding numerical endpoint response is opened. The theory must then survive the native network calculation with that direction unchanged. A successful result therefore tests whether a high-dimensional nuclear network contains a lower-dimensional geometric organization that is shared across reactions, baryon-density anchors, and independent implementations. Within the declared light-sector domain, the three ordered transport directions together with one charge-constrained polarization direction form an exact rank-four reaction-source coordinate. The purpose of the network calculation is not to discover that coordinate by regression, but to test the physical pushforward of a coordinate already fixed by the geometry and conservation laws.

The practical consequence is scalability. If each reaction required its own independently fitted geometric coefficient, the formulation would have little advantage over a table of empirical response factors. Here the same scalar law, baryon-conserving incidence structure, polarization constraint, and baryon-density dependence are carried across the reaction set. Only the reaction stoichiometry and the network-native dynamical pushforward change. This creates a compact translation layer between geometric source structure and conventional nuclear-network evolution: the native rates, thermodynamics, and coupled abundance equations remain authoritative, while CPTG supplies the common source geometry used to organize their response. The resulting operator structure is also computationally useful because once a native susceptibility is established, small source perturbations can be propagated without reconstructing a separate phenomenological model for every channel.

The commissioning paper treated the specific capture channel  $D(p, \gamma)^3\text{He}$  and its reverse photodisintegration channel, carrying a fixed geometric source through a coherent amplitude, an astrophysical  $S$ -factor comparison, a thermonuclear-rate interface, and a separately locked mass-seven transport law [3, 4]. It established that the geometric source could be connected to conventional nuclear observables. The universal construction developed here treats that commissioned channel as one resolved projection of the wider light-element operator rather than as the definition of that operator.

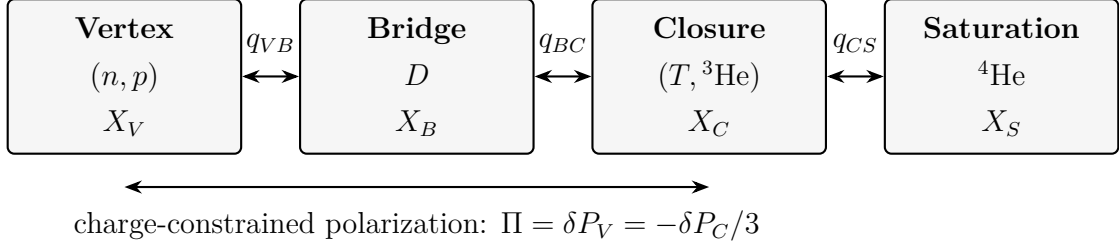
The numerical tests are designed around that distinction. PRIMAT 0.3.2 is used as the primary native authority. A completed full-network campaign spans all 428 native reactions at six baryon-density anchors, and a separately frozen source/Jacobian campaign tests reaction-source predictions without fitting the perturbed endpoints. All 48/48 primary resolved reaction–anchor predictions pass. A prospective full-coordinate PRIMAT test validates the locked mass-seven survival law at the jointly transported CPTG baryon and radiation comparison coordinate in the evolving  $^7\text{Li}$  and  $^7\text{Be}$  channels. PArthENoPE supplies the second independent authority within the shared-endpoint Reaction-20 test. PRyMordial supplies a third completed native authority lane through a 63-reaction, six-anchor, eight-branch campaign whose Reaction-20 endpoint response independently reproduces the PRIMAT and PArthENoPE directions. The aim of these tests is not to force different codes to share an identical numerical susceptibility matrix; it is to test whether the fixed source geometry and its network-native observable pushforward remain consistent when each network is allowed to express its own native dynamics.

A downstream PRIMAT-anchored reduced-graph program probes how far the same transport logic can be followed beyond the exact four-sector theorem. It establishes gap-free modeled support through  $A = 119$  and a numerically qualified reachable frontier at  $A = 338$ , with  $A = 339$  classified as a disconnected rate island of the selected graph. These calculations are included as scalability diagnostics only. They are not native heavy-element PRIMAT evolution, precision primordial heavy-element yield predictions, or a proof that the exact light-sector rank-four theorem extends unchanged to heavy nuclei. The preregistered source-tail robustness test passes all 18/18 retained-source cases, including all 12/12 hard-gated cases.

The paper therefore addresses two linked questions. The first is structural: whether primordial light-nuclear reactions admit a fixed baryon-conserving transport–polarization geometry with a reaction-source space determined before numerical response data are

examined. The second is physical: whether that source geometry is actually carried by authoritative native reaction networks into the native-network endpoint abundance directions and magnitudes. The sections that follow derive the four-sector state and its rank-four source structure, relate it to the commissioned reaction, test the locked mass-seven transport law, construct the network-native pushforward, and then separate the completed native light-sector results from the downstream reachability diagnostics.

### CPTG four-sector light-nuclide geometry



Three ordered transport directions plus one independent polarization direction

Figure 1: Four-sector CPTG light-nuclide geometry. The ordered edge coordinates describe transport from the free-nucleon vertex through the deuterium bridge and mass-three closure to helium-four saturation. The independent coordinate  $\Pi$  retains the charge-polarization information removed by coarse-graining the mirror sectors.

Table 1 separates the fixed analytic inputs, exact geometric statements, physically validated response laws, and implementation-rigidity tests.

| Layer                     | Object                                                                         | Validation status                                                                                                                                               |
|---------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commissioning channel     | $D(p, \gamma){}^3\text{He}$ source, amplitude, $S$ -factor, and rate interface | Established projection test                                                                                                                                     |
| Universal scalar law      | $J_{\text{CPTG}}(\eta_{10})$ and $K_{\text{CPTG}}(\eta_{10})$                  | Fixed analytic input; no standalone scalar-error metric is claimed here                                                                                         |
| Universal state geometry  | Four-sector baryon simplex and ordered edge operator                           | Exact for baryon-conserving tangents                                                                                                                            |
| Reaction-source geometry  | Three transport edges plus one polarization mode                               | Exact rank-four source space                                                                                                                                    |
| Mass-seven live transport | Locked $\tau_7 = \ln \pi$ sink on live ${}^7\text{Li}$ and ${}^7\text{Be}$     | Full-coordinate PRIMAT-native validation complete; 23/23 preregistered scientific gates pass                                                                    |
| Cross-network transfer    | Network-native realization of the fixed rank-four source geometry              | PRIMAT primary validation complete; PArthENoPE second-authority transfer complete; PRyMordial 63-reaction native authority and Reaction-20 replication complete |
| Implementation rigidity   | Matrix-exponential stress and adversarial fault detection                      | Completed within the declared numerical domain                                                                                                                  |

Table 1: Separation of analytic identities, commissioned empirical projection, cross-network numerical validation, and implementation-rigidity tests.

## 2 Relation to the Commissioning Paper

The commissioning paper provides the dimensional nuclear-reaction anchor. It demonstrated that a fixed geometric source state could be carried into a coherent amplitude and a conventional nuclear-astrophysics comparison without using the lithium result to define the capture amplitude. Within the universal construction developed here, that commissioned channel is a resolved projection of the broader operator.

The relationship is

$$\begin{array}{c}
 \text{commissioned source state} \\
 \downarrow \\
 D(p, \gamma)^3\text{He amplitude and rate} \\
 \downarrow \\
 \text{one observable projection of the universal network response.}
 \end{array} \tag{3}$$

The commissioning paper’s locked mass-seven gate is a distinct abundance-transport sector of the core CPTG nuclear theory rather than a coordinate used to construct the four-sector susceptibility. The prospective PRIMAT-native test reported below therefore tests an already-fixed law; it does not use the PRIMAT lithium endpoint to define or retune the transport rule.

## 3 Mass-Seven Live-Channel Transport and PRIMAT-Native Validation

Mass seven is represented by the live native coordinates  ${}^7\text{Li}$  and  ${}^7\text{Be}$ . The abundance-sector coordinate relevant to the locked transport law is

$$Y_7 = Y_{7\text{Li}} + Y_{7\text{Be}}. \tag{4}$$

The CPTG live-channel rule acts on both evolving species [4],

$$\left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{CPTG}} = \left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{native}} - \Gamma_7(t)Y_{7\text{Li}}, \tag{5}$$

$$\left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{CPTG}} = \left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{native}} - \Gamma_7(t)Y_{7\text{Be}}. \tag{6}$$

The integrated optical depth is locked independently of the network endpoint,

$$\tau_7 = \int \Gamma_7(t) dt = \ln \pi, \quad f_7 = e^{-\tau_7} = \frac{1}{\pi}. \tag{7}$$

This law is part of the CPTG nuclear-reaction framework, but it does not introduce a fifth simplex coordinate. The four-sector susceptibility describes the free-nucleon–deuterium–

mass-three-helium-four transport backbone; Equation (6) is a separate abundance-transport action on the live mass-seven channels.

The prospective validation evaluates the locked live-channel law at the complete transported CPTG primordial comparison tuple

$$\Theta_{\text{BBN}}^{\text{CPTG}} = \left( \eta_{10}^{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}}, \tau_7 \right), \quad (8)$$

with

$$\eta_{10}^{\text{BBN}} = 5.998071834744, \quad N_{\text{eff}}^{\text{CPTG}} = 2.968584073464, \quad \tau_7 = \ln \pi. \quad (9)$$

The radiation translation is determined prospectively from two native gate-off calibration rows using only PRIMAT's reported  $N_{\text{eff}}$  response; no deuterium, helium, or lithium abundance enters that mapping. The calibrated slope is

$$\frac{dN_{\text{eff}}}{d(\Delta N_{\text{eff}})} = 0.9999999999999343, \quad (10)$$

and the target input

$$\Delta N_{\text{eff}}^{\text{CPTG}} = -0.07539322509392317 \quad (11)$$

is frozen before target-row abundance exposure. At the transported baryon coordinate, the native gate-off target returns

$$N_{\text{eff}} = 2.968584073463995, \quad (12)$$

an absolute mapping-closure residual of  $4.88 \times 10^{-15}$  relative to the frozen CPTG numerical target. This residual verifies the frozen implementation of the comparison coordinate inside PRIMAT; it is not an independent PRIMAT derivation of the CPTG radiation coordinate.

With the frozen live mass-seven transport enabled, the internal survival factor is

$$\frac{Y_{7,\text{gated}}}{Y_{7,\text{baseline}}} = 0.318366498376680, \quad (13)$$

with relative deviation

$$\left| \frac{(Y_{7,\text{gated}}/Y_{7,\text{baseline}}) - (1/\pi)}{1/\pi} \right| = 1.77852 \times 10^{-4} = 0.0177852\%. \quad (14)$$

PRIMAT's reported  ${}^7\text{Li}/H$  survival factor is 0.318366498376685, giving the same relative

agreement with  $1/\pi$  to the displayed precision. The gated native outputs include

$$D/H_{\text{gated}} = 2.5066925456419876 \times 10^{-5}, \quad (15)$$

$$Y_{p,\text{gated}}^{\text{BBN}} = 0.24576024693245754, \quad (16)$$

$$(^3\text{He}/H)_{\text{gated}} = 1.0503519364716970 \times 10^{-5}, \quad (17)$$

$$(^7\text{Li}/H)_{\text{gated}} = 1.6789263562731326 \times 10^{-10}. \quad (18)$$

The corresponding gate-on/off disturbances of the non-mass-seven controls are

$$|\Delta(D/H)|_{\text{rel}} = 3.91 \times 10^{-9}, \quad (19)$$

$$|\Delta Y_p|_{\text{rel}} = 3.86 \times 10^{-12}, \quad (20)$$

$$|\Delta(^3\text{He}/H)|_{\text{rel}} = 5.59 \times 10^{-10}, \quad (21)$$

$$|\Delta N_{\text{eff}}|_{\text{rel}} = 0. \quad (22)$$

All 23/23 preregistered scientific gates pass. Observational lithium remains report-only and is excluded from every hard acceptance condition; at the declared comparison anchor, the native  $^7\text{Li}/H$  result moves from  $+15.2943\sigma$  gate-off to  $+0.915705\sigma$  gate-on. The result is a prospective PRIMAT-network-native validation of the fixed  $\ln \pi$  live transport law at the jointly transported CPTG baryon and radiation comparison coordinate, not a fitted lithium depletion coefficient, a post-network division by  $\pi$ , or a validation over arbitrary untested cosmological backgrounds [4, 2].

## 4 Inheritance from the Parent CPTG Framework

The universal nuclear-reaction construction does not begin from an independently fitted nuclear geometric parameter. Its comparison coordinates originate in the parent CPTG framework and were introduced in the published *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates* construction [2, 1]. In particular, the parent construction defines the comparison coordinates  $p_C = \pi$  and  $p_{\text{ac}} = 3 - \pi/100$ , together with the transport ratio  $G_T = p_{\text{ac}}/p_C$ , evaluated explicitly in Equation (46). These quantities were fixed before their present use in the four-sector nuclear-response law.

The cross-scale connection is therefore mathematical rather than terminological. The same fixed comparison structure that enters the parent gravitational theory determines the transported primordial coordinate in Equation (26) and the first- and second-order scalar responses in Equations (44) and (45). It is carried into nuclear-reaction space without replacement by an unrelated reaction-specific geometric fit.

The inherited CPTG structure does not replace conventional nuclear physics. Reaction stoichiometry, thermonuclear rates, conservation constraints, and coupled-network propagation remain nuclear inputs. The construction supplies the domain-specific four-sector baryon simplex, transport-polarization source geometry, and source-kernel-normalized

susceptibility through which the inherited scalar structure acts. The resulting construction is therefore a constrained extension of the parent CPTG geometry into atomic and nuclear reaction space, rather than an analogy between the two domains.

## 5 Notation and Perturbation Coordinate

The baryon-to-photon coordinate is

$$\eta_{10} = 10^{10} \frac{n_b}{n_\gamma}, \quad (23)$$

where  $n_b$  and  $n_\gamma$  are the baryon and photon number densities. Species abundances are measured per baryon:

$$Y_i = \frac{n_i}{n_b}. \quad (24)$$

Three baryon-to-photon coordinate roles are kept distinct. The fixed acoustic reference coordinate is

$$\eta_{10}^{\text{ac}} = 6.170380167710, \quad (25)$$

and the transported CPTG abundance coordinate is

$$\eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{G_T} \simeq 5.998071834744. \quad (26)$$

These coordinates follow the fixed pi-branch transport map [2, 1, 4]. The displayed acoustic value is the rounded comparison-coordinate representation used in the pi-branch register; the retained transported value is evaluated from the stored-precision coordinate, so reconstruction from the printed acoustic decimal can differ in the last displayed digits. Elsewhere in this article, an unadorned  $\eta_{10}$  denotes the local response-grid coordinate at which the reaction operator is evaluated. The six validation anchors and the illustrative observational projection at  $\eta_{10} = 6.104$  are response-grid evaluations; they are not identified with the locked transported coordinate in Equation (26).

For a multiplicative perturbation  $f_r$  of reaction  $r$ , the dimensionless symmetric response coordinate is

$$x = -\ln f_r. \quad (27)$$

Thus reciprocal rate perturbations have equal and opposite values of  $x$ . This convention fixes the signs of the first- and second-order coefficients used throughout the paper.

## 6 The Four-Sector Baryon Simplex

At a late BBN endpoint, where the surviving free-neutron abundance is negligible and  $Y_H = Y_p^{\text{free}}$ , define the baryon-normalized coarse-grained state

$$\mathbf{X} = \begin{pmatrix} X_V \\ X_B \\ X_C \\ X_S \end{pmatrix} = \begin{pmatrix} Y_H \\ 2Y_D \\ 3(Y_T + Y_{3\text{He}}) \\ 4Y_{4\text{He}} \end{pmatrix}. \quad (28)$$

The subscripts  $V, B, C, S$  denote vertex, bridge, closure, and saturation.

The baryon sum is

$$\mathbf{1}^T \mathbf{X} = 1 - \epsilon_{A>4}. \quad (29)$$

Using the six unperturbed PRIMAT baseline endpoints at the baryon-density anchors declared in Equation (103), the maximum omitted  $A > 4$  baryon fraction is

$$\max |\epsilon_{A>4}| = 2.8937281459917585 \times 10^{-9}. \quad (30)$$

The corresponding four-sector sum lies in the interval

$$0.9999999971062719 \leq \mathbf{1}^T \mathbf{X} \leq 0.9999999971933646. \quad (31)$$

The complete PRIMAT abundance vectors conserve the baryon-weighted sum to within approximately  $1.7 \times 10^{-12}$  of unity across the same six baseline endpoints. The four-sector truncation is therefore closed to substantially better than one part in  $10^8$ , with the residual explicitly carried by heavier nuclides rather than absorbed into the light-sector coordinates.

Two useful normalized internal coordinates are

$$C_3 = \frac{X_C}{X_B + X_C}, \quad (32)$$

for the bridge-to-closure partition, and

$$S_4 = \frac{X_S}{X_B + X_C + X_S}, \quad (33)$$

for saturation inside the bonded sector. Across the six unperturbed PRIMAT baseline endpoints,

$$0.3864081284244039 \leq C_3 \leq 0.3901312693180608, \quad (34)$$

while

$$0.9997515952667696 \leq S_4 \leq 0.9997562469639383. \quad (35)$$

The bonded population is therefore overwhelmingly saturated into helium-four, while the residual bridge/closure partition remains a sensitive diagnostic of network response.

## 7 Exact Ordered Edge Transport

The ordered state path is represented by the incidence matrix

$$\mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}. \quad (36)$$

For edge-current vector

$$\mathbf{j} = \begin{pmatrix} j_{VB} \\ j_{BC} \\ j_{CS} \end{pmatrix}, \quad (37)$$

the state response is

$$\frac{d\mathbf{X}}{dx} = \mathbf{D}\mathbf{j}. \quad (38)$$

Because

$$\mathbf{1}^T \mathbf{D} = 0, \quad (39)$$

the operator preserves the baryon simplex exactly.

For any zero-sum state tangent

$$\mathbf{v} = (v_V, v_B, v_C, v_S)^T, \quad (40)$$

the edge decomposition is unique. The vertex-to-bridge current is

$$j_{VB} = -v_V. \quad (41)$$

The bridge-to-closure current is

$$j_{BC} = -(v_V + v_B). \quad (42)$$

The closure-to-saturation current is

$$j_{CS} = v_S. \quad (43)$$

Equation (38) is exact for zero-sum tangents. When the four-sector truncation is compared with a complete PRIMAT endpoint, the leading closure mismatch is the explicitly measured  $A > 4$  baryon fraction in Equation (30); it is not an error of the incidence operator.

## 8 Scalar Curvature Response Law

The fixed CPTG first-order geometric response is

$$J_{\text{CPTG}}(\eta_{10}) = \frac{1}{\sqrt{8}} \sqrt{\frac{\eta_{10}}{\eta_{10}^{\text{ac}}}}, \quad (44)$$

with the acoustic reference coordinate defined in Equation (25). The pi-branch construction fixes both this reference coordinate and the transport ratio used below [2, 1].

The fixed CPTG second-order curvature coefficient is

$$K_{\text{CPTG}}(\eta_{10}) = -\frac{3}{16} \frac{\eta_{10}}{\eta_{10}^{\text{ac}}} G_T^{-1/3}, \quad (45)$$

where

$$G_T = \frac{3 - \pi/100}{\pi} = 0.9449296585513721. \quad (46)$$

Here  $G_T = p_{\text{ac}}/p_C$ , with  $p_C = \pi$  and  $p_{\text{ac}} = 3 - \pi/100$ , as defined by the CPTG geometric pi branch [2, 1].

Equivalently,

$$K_{\text{CPTG}} = -\frac{3}{2} J_{\text{CPTG}}^2 G_T^{-1/3}. \quad (47)$$

A compact generating function reproducing the fixed first and second coefficients is

$$g_T(x; \eta_{10}) = \frac{s_T}{3} \ln \left( 1 + \frac{3J_{\text{CPTG}}}{s_T} x \right), \quad s_T = G_T^{1/3}. \quad (48)$$

Its expansion begins

$$g_T(x) = Jx - \frac{3J^2}{2s_T} x^2 + \frac{3J^3}{s_T^2} x^3 + \mathcal{O}(x^4). \quad (49)$$

The first- and second-order coefficients are retained explicitly as fixed coefficients of the present CPTG construction. Higher coefficients remain generator-implied continuations until separately tested.

The logarithm in Equation (48) fixes the admissible perturbative domain:

$$1 + \frac{3J_{\text{CPTG}}(\eta_{10})}{s_T} x > 0. \quad (50)$$

The boundary at equality is a genuine logarithmic singularity. All numerical validation and deployment statements in this paper are restricted to the interior of the domain in Equation (50).

## 9 Scalar-Law Validation Boundary

The scalar coefficients in Equations (44) and (45) remain fixed analytic inputs to the CPTG construction. The completed PRIMAT evidence reported below directly tests the reaction-source geometry and its network-native first-order pushforward; it is not a standalone numerical test of the scalar coefficients themselves. Accordingly, this paper does not quote a scalar-law bias, mean absolute error, root-mean-square error, or improvement factor from the PRIMAT mechanism campaign.

The six baryon-density anchors used by the completed PRIMAT validation are

$$\eta_{10} \in \{6.044, 6.080, 6.094, 6.106, 6.128, 6.134\}. \quad (51)$$

No coefficient in  $J_{\text{CPTG}}$ ,  $K_{\text{CPTG}}$ ,  $G_T$ , or  $\eta_{10}^{\text{ac}}$  is adjusted in response to the PRIMAT endpoint measurements. The quantitative PRIMAT claim in this article is the completed native source/Jacobian comparison in Section 14.3, where the fixed reaction-source geometry is propagated through PRIMAT's native dynamics before comparison with independently measured first-order endpoint susceptibilities.

## 10 Dynamic Free-Nucleon Vertex

The late endpoint definition  $X_V = Y_H$  must be extended during the active reaction epoch because the free-neutron population is not negligible. The exact dynamic vertex is

$$X_V = Y_n + Y_p^{\text{free}}. \quad (52)$$

The complete dynamic state is

$$\mathbf{X} = \begin{pmatrix} Y_n + Y_p^{\text{free}} \\ 2Y_D \\ 3(Y_T + Y_{3\text{He}}) \\ 4Y_{4\text{He}} \end{pmatrix}. \quad (53)$$

As  $Y_n \rightarrow 0$ , the dynamic definition reduces to the late endpoint coordinate.

The grouped states retain internal charge orientation through

$$P_V = Y_p^{\text{free}} - Y_n, \quad P_C = 3(Y_{3\text{He}} - Y_T). \quad (54)$$

The six species are reconstructed exactly. The free-neutron abundance is

$$Y_n = \frac{X_V - P_V}{2}. \quad (55)$$

The free-proton abundance is

$$Y_p^{\text{free}} = \frac{X_V + P_V}{2}. \quad (56)$$

The deuterium abundance is

$$Y_D = \frac{X_B}{2}. \quad (57)$$

The tritium abundance is

$$Y_T = \frac{X_C - P_C}{6}. \quad (58)$$

The helium-three abundance is

$$Y_{\text{He}^3} = \frac{X_C + P_C}{6}. \quad (59)$$

The helium-four abundance is

$$Y_{\text{He}^4} = \frac{X_S}{4}. \quad (60)$$

For the ordered map

$$(Y_n, Y_p^{\text{free}}, Y_D, Y_T, Y_{\text{He}^3}, Y_{\text{He}^4}) \longrightarrow (X_V, P_V, X_B, X_C, P_C, X_S), \quad (61)$$

the transformation determinant is 288, so the coordinate map is invertible.

## 11 Charge-Constrained Polarization

The total baryon and charge coordinates are

$$B = X_V + X_B + X_C + X_S \quad (62)$$

and

$$Q = \frac{B}{2} + \frac{P_V}{2} + \frac{P_C}{6}. \quad (63)$$

For every baryon- and charge-conserving reaction perturbation within the defined six-species light sector,

$$3\delta P_V + \delta P_C = 0. \quad (64)$$

Therefore only one independent internal polarization mode remains. Define

$$\Pi = \delta P_V = -\frac{\delta P_C}{3}. \quad (65)$$

The universal reaction-source vector is

$$\mathbf{s}_r = \begin{pmatrix} q_{VB} \\ q_{BC} \\ q_{CS} \\ \Pi \end{pmatrix}_r. \quad (66)$$

Within the defined coordinate spanning free nucleons, deuterium, mass three, and helium four, the baryon- and charge-conserving reaction-source tangent space is four-dimensional. Its basis contains three ordered transport edges and one charge-constrained polarization direction.

The conventional abundance ratios are projections of this extended coordinate. For example,

$$\frac{D}{H} = \frac{X_B}{X_V + P_V}, \quad (67)$$

and

$$\frac{{}^3\text{He}}{H} = \frac{X_C + P_C}{3(X_V + P_V)}. \quad (68)$$

Thus the commissioned  $D/H$  law is a mixed transport-polarization projection, not the complete universal matrix.

## 12 Exact Reaction-Source Basis

The exact reaction stoichiometry generates integer source vectors in the transport–polarization basis  $(q_{VB}, q_{BC}, q_{CS}, \Pi)$ . Four useful directions are:

$$\mathbf{s}_{12} = (2, 0, 0, 0)^T. \quad (69)$$

$$\mathbf{s}_{16} = (0, 0, 0, 2)^T. \quad (70)$$

$$\mathbf{s}_{28} = (-1, 3, 0, -1)^T. \quad (71)$$

$$\mathbf{s}_{30} = (-1, 1, 4, -1)^T. \quad (72)$$

The numerical reaction identifiers are tied to explicit physical channels in Table 2; they are not standalone theoretical labels. The table also publishes every source vector used by the construction, independent validation, and structural diagnostic sets.

Table 2: Physical reaction mapping and complete transport-polarization source-vector register.

| ID | Physical channel                                                   | $\mathbf{s}_r = (q_{VB}, q_{BC}, q_{CS}, \Pi)^T$ | Validation role        |
|----|--------------------------------------------------------------------|--------------------------------------------------|------------------------|
| 12 | $n + p \leftrightarrow D + \gamma$                                 | $(2, 0, 0, 0)^T$                                 | Construction           |
| 13 | $D + n \leftrightarrow T + \gamma$                                 | $(1, 3, 0, 1)^T$                                 | Structural diagnostic  |
| 14 | ${}^3\text{He} + n \leftrightarrow {}^4\text{He} + \gamma$         | $(1, 1, 4, 1)^T$                                 | Independent validation |
| 16 | ${}^3\text{He} + n \leftrightarrow p + T$                          | $(0, 0, 0, 2)^T$                                 | Construction           |
| 20 | $D + p \leftrightarrow {}^3\text{He} + \gamma$                     | $(1, 3, 0, -1)^T$                                | Independent validation |
| 21 | $T + p \leftrightarrow {}^4\text{He} + \gamma$                     | $(1, 1, 4, -1)^T$                                | Structural diagnostic  |
| 28 | $D + D \leftrightarrow n + {}^3\text{He}$                          | $(-1, 3, 0, -1)^T$                               | Construction           |
| 29 | $D + D \leftrightarrow p + T$                                      | $(-1, 3, 0, 1)^T$                                | Independent validation |
| 30 | $T + D \leftrightarrow n + {}^4\text{He}$                          | $(-1, 1, 4, -1)^T$                               | Construction           |
| 31 | ${}^3\text{He} + D \leftrightarrow p + {}^4\text{He}$              | $(-1, 1, 4, 1)^T$                                | Structural diagnostic  |
| 32 | ${}^3\text{He} + {}^3\text{He} \leftrightarrow 2p + {}^4\text{He}$ | $(-2, -2, 4, 2)^T$                               | Structural diagnostic  |
| 38 | ${}^3\text{He} + T \leftrightarrow D + {}^4\text{He}$              | $(0, -2, 4, 0)^T$                                | Structural diagnostic  |
| 39 | $T + T \leftrightarrow 2n + {}^4\text{He}$                         | $(-2, -2, 4, -2)^T$                              | Independent validation |
| 40 | ${}^3\text{He} + T \leftrightarrow n + p + {}^4\text{He}$          | $(-2, -2, 4, 0)^T$                               | Structural diagnostic  |

These form the fixed construction basis. Its determinant is

$$\det \mathbf{S}_D = 48, \quad (73)$$

and its column-normalized two-norm condition number is

$$\kappa_2(\mathbf{S}_D) = 1.7402818881. \quad (74)$$

The pure axes are reconstructed exactly:

$$\mathbf{e}_{VB} = \frac{1}{2}\mathbf{s}_{12}. \quad (75)$$

$$\mathbf{e}_{\Pi} = \frac{1}{2}\mathbf{s}_{16}. \quad (76)$$

$$\mathbf{e}_{BC} = \frac{1}{6}\mathbf{s}_{12} + \frac{1}{6}\mathbf{s}_{16} + \frac{1}{3}\mathbf{s}_{28}. \quad (77)$$

$$\mathbf{e}_{CS} = \frac{1}{12}\mathbf{s}_{12} + \frac{1}{12}\mathbf{s}_{16} - \frac{1}{12}\mathbf{s}_{28} + \frac{1}{4}\mathbf{s}_{30}. \quad (78)$$

The fixed held-out zero-refit validation basis is

$$(14, 20, 29, 39), \quad (79)$$

with determinant

$$\det \mathbf{S}_H = -144 \quad (80)$$

and column-normalized condition number

$$\kappa_2(\mathbf{S}_H) = 2.3926748723. \quad (81)$$

This held-out zero-refit validation basis spans the full source space. Reactions 14, 20, 29, and 39 provide independent light-element tests of the same fixed CPTG source construction. Their measured endpoint responses do not define the reaction-source vectors, alter the geometric basis, or authorize reaction-specific corrections.

Table 3 summarizes the full-rank separation between model construction and independent testing.

| Set                            | Reactions              | Determinant          | Role                                                                                                                                 |
|--------------------------------|------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Geometric construction         | 12, 16, 28, 30         | +48                  | Provides a full-rank basis for reconstructing each network-native realization of the universal source geometry                       |
| Held-out zero-refit validation | 14, 20, 29, 39         | −144                 | Tests all four source directions with the fixed CPTG construction; no validation reaction modifies the source geometry or scalar law |
| Structural diagnostic          | 13, 21, 31, 32, 38, 40 | not used for fitting | Mirror-polarization and mixed-direction diagnostics                                                                                  |

Table 3: Fixed full-rank light-element channel design for construction and held-out zero-refit testing.

### 13 Network-Native Observable Pushforward

The exact CPTG source geometry is universal within the declared four-sector domain, but the numerical realization of a source perturbation is network-local. For a network implementation  $\mathcal{N}$ , let  $\mathcal{F}_{r,\mathcal{N}}(x)$  denote a code-local source-kernel coordinate for reaction  $r$ . Its local expansion is

$$\mathcal{F}_{r,\mathcal{N}}(x) = \mathcal{A}_{r,\mathcal{N}}^{(1)}x + \mathcal{B}_{r,\mathcal{N}}^{(2)}x^2 + \mathcal{O}(x^3). \quad (82)$$

The corresponding order-resolved source drives are

$$\mathbf{d}_{r,\mathcal{N}}^{(1)} = \mathcal{A}_{r,\mathcal{N}}^{(1)}\mathbf{s}_r, \quad \mathbf{d}_{r,\mathcal{N}}^{(2)} = \mathcal{B}_{r,\mathcal{N}}^{(2)}\mathbf{s}_r. \quad (83)$$

Let  $\mathbf{y}_{r,\mathcal{N}}^{(1)}$  denote the first-order response vector in a declared set of observables defined within network  $\mathcal{N}$ . The validated network-native pushforward is

$$\mathbf{y}_{r,\mathcal{N}}^{(1)} = \mathbf{L}_{\mathcal{N}}^{(1)}(\eta_{10})\mathbf{d}_{r,\mathcal{N}}^{(1)}. \quad (84)$$

The completed native source/Jacobian mechanism test concerns this first-order map. The quadratic endpoint coefficient  $c_2$  extracted from the symmetric reaction ladders remains a network-native second-order diagnostic, and  $K_{\text{CPTG}}$  remains a fixed scalar coefficient of the CPTG generator. In a nonlinear reaction network, however, the complete second

variation generally contains both direct second-order source forcing and terms bilinear in the first-order state/source through the curvature of the native dynamics. Accordingly, this paper does not assert a closed relation of the form  $\mathbf{y}^{(2)} = \mathbf{L}^{(2)}\mathbf{d}^{(2)}$  and does not claim a validated second-order source/Jacobian susceptibility matrix.

The dimension and numerical entries of  $\mathbf{L}_{\mathcal{N}}^{(1)}$  depend on the declared observable set and on the native network dynamics. They are not assumed to be identical between codes. The universal claim concerns the fixed source coordinate  $\mathbf{s}_r$ , its rank-four transport-polarization structure, the scalar CPTG law, and the validated first-order source-to-observable factorization; it does not require equality of code-local current amplitudes, Jacobians, source kernels, integration measures, or solver trajectories.

For the shared endpoint comparisons, define the projection

$$\mathbf{P}_3\mathbf{y} = \begin{pmatrix} \Delta \ln(D/H) \\ \Delta \ln(^3\text{He}/H) \\ \Delta \ln(^7\text{Li}/H) \end{pmatrix}. \quad (85)$$

This projection is common to the declared PRIMAT, PARthENoPE, and PRyMordial Reaction-20 endpoint comparisons and provides a shared observable space without asserting a universal numerical susceptibility matrix. The vector in this observable space is a network-native endpoint response produced after dynamical pushforward; it is not identified directly with the bare geometric source vector.

PRIMAT supplies the primary native source-level realization: for each mapped reaction, the isolated PRIMAT reaction source is propagated through the native analytic Jacobian to the endpoint with independent BDF and Radau variational integrations. All 84 reaction-anchor predictions are frozen before the independently measured endpoint coefficients are opened, so the endpoint responses do not construct or fit the source-level prediction. PARthENoPE supplies a separate completed full-network Reaction-20 transfer. PRyMordial supplies a completed 63-reaction full-native response surface and a third independent Reaction-20 endpoint replication.

## 14 Empirical and Cross-Network Validation

The construction is tested at three distinct levels: direct comparison of the commissioned reaction with nuclear data, cross-network response transfer, and completed native-mechanism validation in the primary network. These tests constrain different parts of the theory and are therefore reported separately.

### 14.1 Commissioning-Channel Empirical Projection

The commissioned  $D(p, \gamma)^3\text{He}$  projection was compared with the direct measurements of Tišma, Turkat, and Dhibar and with the thermonuclear-rate evaluation of Moscoso et

al. [12, 13, 14, 15]. The Mossa/LUNA data were replayed directly on the published 13-point  $S$ -factor grid from 32.4 to 262.9 keV and on the associated 17-point thermonuclear-rate table from 0.001 to 10 GK [16]. Tables 4 and 5 report the complete point-by-point replay.

## Complete Mossa/LUNA Point-by-Point Replay Tables

For the direct  $S$ -factor comparison, the statistical and systematic uncertainties are combined in quadrature,

$$\sigma_{S,i} = \sqrt{\sigma_{\text{stat},i}^2 + \sigma_{\text{sys},i}^2}, \quad z_{S,i} = \frac{S_{\text{CPTG},i} - S_{\text{obs},i}}{\sigma_{S,i}}. \quad (86)$$

All 13 of 13 points satisfy  $|z_{S,i}| < 1$ , giving an explicit 13/13 pointwise result. At full calculation precision, the diagonal pointwise sum of squared pulls is  $\sum_i z_{S,i}^2 = 2.563667670328492$ , with mean squared pull 0.197205205409884, and the largest absolute pull is 0.951713181055002 at 132.9 keV. This diagnostic uses the listed pointwise uncertainties and is not a covariance-aware global likelihood. Thus,

$$N_{|z_S|<1} = \frac{13}{13}. \quad (87)$$

| $E$ (keV) | $S_{\text{obs}}$ (eV b) | $\sigma_{\text{stat}}/\sigma_{\text{sys}}$ (eV b) | $\sigma_{S,\text{comb}}$ (eV b) | $S_{\text{CPTG}}$ (eV b) | $\Delta S = S_{\text{CPTG}} - S_{\text{obs}}$ (eV b) | $z_S$     |
|-----------|-------------------------|---------------------------------------------------|---------------------------------|--------------------------|------------------------------------------------------|-----------|
| 32.4      | 0.386                   | 0.014 / 0.010                                     | 0.017204651                     | 0.385664125              | -0.000335875                                         | -0.019522 |
| 66.7      | 0.627                   | 0.009 / 0.016                                     | 0.018357560                     | 0.621787119              | -0.005212881                                         | -0.283964 |
| 99.5      | 0.850                   | 0.008 / 0.021                                     | 0.022472205                     | 0.856281226              | +0.006281226                                         | +0.279511 |
| 115.9     | 0.966                   | 0.009 / 0.024                                     | 0.025632011                     | 0.976527265              | +0.010527265                                         | +0.410708 |
| 132.9     | 1.133                   | 0.004 / 0.031                                     | 0.031256999                     | 1.103252302              | -0.029747698                                         | -0.951713 |
| 149.3     | 1.223                   | 0.006 / 0.031                                     | 0.031575307                     | 1.227488811              | +0.004488811                                         | +0.142162 |
| 166.1     | 1.375                   | 0.004 / 0.036                                     | 0.036221541                     | 1.356753008              | -0.018246992                                         | -0.503761 |
| 182.7     | 1.475                   | 0.006 / 0.037                                     | 0.037483330                     | 1.486441485              | +0.011441485                                         | +0.305242 |
| 199.5     | 1.648                   | 0.003 / 0.043                                     | 0.043104524                     | 1.619656314              | -0.028343686                                         | -0.657557 |
| 222.8     | 1.791                   | 0.006 / 0.045                                     | 0.045398238                     | 1.807639449              | +0.016639449                                         | +0.366522 |
| 232.9     | 1.866                   | 0.012 / 0.051                                     | 0.052392748                     | 1.890276957              | +0.024276957                                         | +0.463365 |
| 252.9     | 2.073                   | 0.012 / 0.052                                     | 0.053366656                     | 2.055943869              | -0.017056131                                         | -0.319603 |
| 262.9     | 2.156                   | 0.020 / 0.054                                     | 0.057584720                     | 2.139778294              | -0.016221706                                         | -0.281702 |

Table 4: Complete 13/13 replay of the Mossa/LUNA astrophysical  $S$ -factor measurements. The uncertainty-pair column lists the published statistical and systematic uncertainties, respectively;  $\sigma_S$  and  $z_S$  are defined by Equation (86). All 13 CPTG predictions lie within the combined one-standard-deviation interval. Displayed values are rounded from the full-precision calculation used for the summary statistics.

The thermonuclear-rate table reports asymmetric published one-standard-deviation bounds  $R_i^-$  and  $R_i^+$ . The acceptance decision is direct interval inclusion,

$$R_i^- \leq R_{\text{CPTG},i} \leq R_i^+. \quad (88)$$

For a signed scalar diagnostic, without replacing the published asymmetric interval by a symmetric Gaussian likelihood, the table also gives the recorded half-band-normalized residual

$$\tilde{z}_{R,i} = \frac{R_{\text{CPTG},i} - R_{\text{obs},i}}{(R_i^+ - R_i^-)/2}. \quad (89)$$

All 17 of 17 rates satisfy Equation (88), giving an explicit 17/17 interval result. The largest absolute band-normalized residual is 0.959051449382153 at  $T_9 = 10$ . Thus,

$$N_{R^- \leq R_{\text{CPTG}} \leq R^+} = \frac{17}{17}. \quad (90)$$

| $T_9$  | $R_{\text{obs}}$       | $R^-$                  | $R^+$                  | $R_{\text{CPTG}}$            | $\Delta_R$ (%) | $\tilde{z}_R$ |
|--------|------------------------|------------------------|------------------------|------------------------------|----------------|---------------|
| 0.001  | $1.37 \times 10^{-11}$ | $1.35 \times 10^{-11}$ | $1.39 \times 10^{-11}$ | $1.37549885 \times 10^{-11}$ | +0.401376      | +0.274942     |
| 0.005  | $2.57 \times 10^{-5}$  | $2.53 \times 10^{-5}$  | $2.62 \times 10^{-5}$  | $2.59614454 \times 10^{-5}$  | +1.017297      | +0.580990     |
| 0.010  | $1.53 \times 10^{-3}$  | $1.51 \times 10^{-3}$  | $1.56 \times 10^{-3}$  | $1.55277174 \times 10^{-3}$  | +1.488349      | +0.910870     |
| 0.050  | 0.908                  | 0.894                  | 0.922                  | 0.918597756                  | +1.167154      | +0.756983     |
| 0.100  | 5.74                   | 5.65                   | 5.84                   | 5.659904627                  | -1.395390      | -0.843109     |
| 0.500  | 129                    | 126                    | 132                    | 128.358495                   | -0.497290      | -0.213835     |
| 1.000  | 363                    | 352                    | 374                    | 363.206497                   | +0.056886      | +0.018772     |
| 1.500  | 632                    | 609                    | 656                    | 631.152521                   | -0.134095      | -0.036063     |
| 2.000  | 920                    | 879                    | 962                    | 915.950702                   | -0.440141      | -0.097573     |
| 3.000  | 1520                   | 1430                   | 1610                   | 1511.110426                  | -0.584840      | -0.098773     |
| 4.000  | 2110                   | 1950                   | 2280                   | 2118.165937                  | +0.387011      | +0.049491     |
| 5.000  | 2670                   | 2400                   | 2930                   | 2719.253573                  | +1.844703      | +0.185863     |
| 6.000  | 3160                   | 2760                   | 3550                   | 3300.645090                  | +4.450794      | +0.356064     |
| 7.000  | 3560                   | 3000                   | 4120                   | 3851.134650                  | +8.177940      | +0.519883     |
| 8.000  | 3850                   | 3090                   | 4610                   | 4361.958427                  | +13.297621     | +0.673630     |
| 9.000  | 4010                   | 3020                   | 5010                   | 4826.999562                  | +20.374054     | +0.821105     |
| 10.000 | 4020                   | 2750                   | 5300                   | 5242.790598                  | +30.417677     | +0.959051     |

Table 5: Complete 17/17 replay of the Mossa/LUNA thermonuclear reaction-rate entries. Rates are in  $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$ . The published lower and upper columns are the asymmetric one-standard-deviation bounds. Every CPTG value lies within its published interval. Here,  $\Delta_R = 100(R_{\text{CPTG}}/R_{\text{obs}} - 1)$ . The signed  $\tilde{z}_R$  column is the half-band-normalized diagnostic defined by Equation (89), not a replacement symmetric likelihood. Displayed CPTG values are rounded from the full-precision integrations used for the interval tests.

At the 91 keV pivot, the CPTG candidate gives

$$S_{\text{CPTG}}(91 \text{ keV}) = 0.7931440807 \text{ eV b}, \quad (91)$$

compared with the reference value

$$S_{\text{ref}}(91 \text{ keV}) = (0.799 \pm 0.018) \text{ eV b}. \quad (92)$$

The corresponding normalized residual is  $-0.3253$ . Across the 54-point rate grid, 52 points lie within the quoted one-standard-deviation band and all 54 lie within the approximate

two-standard-deviation band. At  $T_9 = 0.8$ , the candidate rate is 263.5475 compared with 264.2 in the reference evaluation, a normalized residual of  $-0.1136$ .

| Dataset | Statistical role                  | $r$ (N) | Fitted, $k$ | $\nu = N - k$ | $\chi^2$ | $\chi^2/\nu$ | $p$ -value             |
|---------|-----------------------------------|---------|-------------|---------------|----------|--------------|------------------------|
| Tišma   | Correlated shape-consistency test | 4       | 0           | 4             | 0.9301   | 0.2325       | 0.9202                 |
| Turkat  | Correlated shape-consistency test | 14      | 0           | 14            | 15.5915  | 1.1137       | 0.3390                 |
| Dhibar  | External pointwise comparison     | 3       | 0           | 3             | 46.5519  | 15.5173      | $4.33 \times 10^{-10}$ |

Table 6: Covariance-weighted comparisons of the fixed CPTG commissioning-channel projection with the Tišma, Turkat, and Dhibar datasets. No normalization, slope, or other dataset-specific parameter was fitted in these comparisons, so the reported nominal degrees of freedom equal the listed residual-vector dimensions.

Table 6 compares three experimental datasets with the same fixed CPTG projection; the different degrees of freedom do not represent different parameter counts in competing models. They arise from the number of residual components supplied by each dataset after covariance weighting. The Tišma and Turkat results show consistency with the projected energy dependence, with reduced chi-square values of 0.2325 and 1.1137, respectively. Because both datasets contributed to the external shape reference, their  $p$ -values are nominal in-sample consistency diagnostics rather than out-of-sample model-selection probabilities. The Dhibar comparison is an external three-point test and is strongly inconsistent as a complete dataset, with its chi-square dominated by the discordant highest-energy point. The bounded empirical conclusion is therefore unchanged: the commissioned CPTG projection agrees with the pivot, rate-grid, Mossa/LUNA, Tišma, and Turkat comparisons, but it does not claim pointwise agreement with every published measurement.

## 14.2 Full-Network PArthENoPE Reaction-20 Transfer and Replication

The CPTG Reaction-20 shared-endpoint response was independently evaluated in the official PArthENoPE 3.0 full network [10, 11]. The CPTG scalar law, reaction-source construction, baryon-density dependence, and observable projection were unchanged, and no PArthENoPE-specific geometric parameter or correction factor was introduced.

For cross-network comparison, the first-order native endpoint vector is

$$\mathbf{c}_{1,\mathcal{N}}^{(20)} = \begin{pmatrix} c_{1,D/H} \\ c_{1,^3\text{He}/H} \\ c_{1,^7\text{Li}/H} \end{pmatrix}_{\mathcal{N}}, \quad (93)$$

with the coefficients extracted from the logarithmic ladder fit in Equation (96). This network-native observable vector follows the source-to-endpoint pushforward and must not be replaced by the bare geometric source coordinate.

| Validation quantity                                       | Measured result   | Requirement  |
|-----------------------------------------------------------|-------------------|--------------|
| Native rows                                               | 695/695           | 695/695      |
| Matched branch pairs                                      | 338/338           | 338/338      |
| Complete eight-branch ladders                             | 84/84             | 84/84        |
| Reaction-20 direction cosine to frozen endpoint reference | 0.999979437501036 | $\geq 0.995$ |
| $D/H$ component ratio                                     | 0.998516625       | 0.85–1.15    |
| $^3\text{He}/H$ component ratio                           | 0.993911268       | 0.85–1.15    |
| $^7\text{Li}/H$ component ratio                           | 1.008589280       | 0.85–1.15    |
| $D/H$ six-anchor variation                                | 2.491%            | $\leq 10\%$  |
| $^3\text{He}/H$ six-anchor variation                      | 0.809%            | $\leq 10\%$  |
| $^7\text{Li}/H$ six-anchor variation                      | 1.156%            | $\leq 10\%$  |

Table 7: Zero-refit full-network PArthENoPE transfer and replication of the fixed Reaction-20 shared-endpoint response.

The 695 rows, 338 matched pairs, and 84 ladders constitute the complete archived PArthENoPE campaign and satisfy their integrity and completeness contracts. The archived direction and component-ratio gates were evaluated against the frozen shared-endpoint reference near  $(+0.343, -0.389, -0.603)$ , not against the bare geometric source coordinate. A direct raw-response comparison with PRIMAT gives a minimum six-anchor cosine of 0.999974569, a median cosine of 0.999981322, and a worst relative vector difference of 0.966%. The median native vectors are

$$\begin{aligned}\bar{\mathbf{c}}_{20}^{\text{PRIMAT}} &= (+0.343898, -0.387450, -0.602684)^T, \\ \bar{\mathbf{c}}_{20}^{\text{PArthENoPE}} &= (+0.342815, -0.386472, -0.608490)^T.\end{aligned}\tag{94}$$

The result therefore establishes zero-refit transfer of the Reaction-20 network-native endpoint response without promoting every reaction in the 695-row campaign to a universal-law pass.

### 14.3 PRIMAT Six-Anchor Native Source/Jacobian Validation

PRIMAT 0.3.2 is the primary validation network for the completed full-reaction campaign [8]. A separate native-mechanism calculation tests whether the fixed CPTG reaction-source geometry is carried by PRIMAT’s own dynamics to the independently measured first-order endpoint susceptibilities. Fourteen mapped light-sector reactions are evaluated at all six baryon-density anchors, giving

$$14 \times 6 = 84\tag{95}$$

first-order variational predictions. The prediction phase uses the isolated native reaction source, PRIMAT’s analytic network Jacobian, and independent BDF and Radau propagation. The complete 84-row prediction table is frozen before the authoritative perturbed-endpoint coefficients are opened for evaluation; no endpoint coefficient is used to fit a variational prediction.

The endpoint reference coefficients are independently extracted from the completed eight-branch reaction ladders using

$$\Delta \ln Y = c_1 x + c_2 x^2, \quad x = -\ln f_r, \quad (96)$$

with zero intercept. For a primary resolved response, defined by  $\|\mathbf{c}_1\| \geq 10^{-3}$ , the frozen acceptance gates are

$$\frac{\|\mathbf{c}_{1,\text{var}} - \mathbf{c}_{1,\text{ref}}\|}{\|\mathbf{c}_{1,\text{ref}}\|} \leq 0.02, \quad \cos \theta \geq 0.999, \quad (97)$$

while the independent BDF/Radau realizations must satisfy

$$\epsilon_{\text{BDF/Radau}} \leq 5 \times 10^{-4}, \quad \cos \theta_{\text{BDF/Radau}} \geq 0.999999. \quad (98)$$

The native source-direction and baryon residuals must each remain below  $10^{-10}$ , and the six unperturbed Python/C baselines must agree within  $5 \times 10^{-6}$ .

| $\eta_{10}$ | Primary rows | Max. vector discrepancy    | Min. direction cosine | Max. BDF/Radau discrepancy |
|-------------|--------------|----------------------------|-----------------------|----------------------------|
| 6.044       | 8/8          | $3.6762871 \times 10^{-4}$ | 0.9999999924210492    | $5.0978860 \times 10^{-7}$ |
| 6.080       | 8/8          | $3.6552796 \times 10^{-4}$ | 0.9999999922770632    | $5.7253922 \times 10^{-7}$ |
| 6.094       | 8/8          | $3.6556010 \times 10^{-4}$ | 0.9999999922394864    | $5.8248229 \times 10^{-7}$ |
| 6.106       | 8/8          | $3.6395324 \times 10^{-4}$ | 0.9999999921946300    | $5.1557128 \times 10^{-7}$ |
| 6.128       | 8/8          | $3.6290749 \times 10^{-4}$ | 0.9999999921069808    | $5.0349449 \times 10^{-7}$ |
| 6.134       | 8/8          | $3.6424119 \times 10^{-4}$ | 0.9999999920939400    | $4.9939331 \times 10^{-7}$ |

Table 8: Completed PRIMAT six-anchor native source/Jacobian validation. Each anchor contributes eight primary resolved reaction tests; all 48/48 primary tests satisfy the frozen endpoint and cross-solver gates.

Across all six anchors, the worst primary endpoint-vector discrepancy is 0.0367629%, the minimum primary direction cosine is 0.99999999209394, and the largest BDF/Radau relative difference is  $5.824823 \times 10^{-7}$ . The source audit contains 6,283 active native source samples; both the source-direction and baryon residuals remain below  $4.44 \times 10^{-16}$ . All 48/48 primary resolved tests pass, and 66/84 reaction-anchor cases lie at or above the  $10^{-4}$  secondary reporting floor. The maximum six-anchor unperturbed Python/C baseline discrepancy is  $2.631735 \times 10^{-6}$ , below the frozen  $5 \times 10^{-6}$  gate.

For the commissioned  $D(p, \gamma)^3\text{He}$  channel, the largest six-anchor endpoint-vector discrepancy across either variational solver is approximately  $1.62 \times 10^{-4}$  (0.0162%), with direction cosine effectively unity. The completed PRIMAT result therefore validates the network-native dynamical pushforward of the fixed reaction-source geometry rather than identifying a bare source vector directly with a final abundance-response vector.

#### 14.4 PRyMordial 63-Reaction Native Authority and Reaction-20 Replication

PRyMordial supplies a third completed native response authority using its 63-reaction network [9]. The campaign uses the same six baryon-density anchors and eight symmetric

logarithmic rate branches employed in the shared response analysis. With six unperturbed baselines, the complete execution count is

$$N_{\text{PRyMordial}} = 6 + 63 \times 6 \times 8 = 3,030. \quad (99)$$

All 3,030/3,030 rows are committed. The completed authority contains all 378/378 reaction–anchor ladders and 1,512/1,512 matched symmetric branch pairs, with no invalid committed rows. Direct native rate-scaling proofs have worst relative error  $4.44 \times 10^{-16}$  over the accepted rows.

The Reaction-20 endpoint coefficients are extracted with the same zero-intercept quadratic logarithmic fit used in Equation (96). The resulting six-anchor vectors are

| $\eta_{10}$ | $c_{1,D/H}$  | $c_{1,^3\text{He}/H}$ | $c_{1,^7\text{Li}/H}$ |
|-------------|--------------|-----------------------|-----------------------|
| 6.044       | +0.339044367 | −0.384827087          | −0.599192799          |
| 6.080       | +0.341740827 | −0.386268722          | −0.600935916          |
| 6.094       | +0.342790644 | −0.386826434          | −0.601605728          |
| 6.106       | +0.343691177 | −0.387303646          | −0.602175734          |
| 6.128       | +0.345342909 | −0.388175962          | −0.603212518          |
| 6.134       | +0.345794400 | −0.388412945          | −0.603492407          |

Table 9: Completed PRyMordial six-anchor Reaction-20 network-native endpoint response. The component order matches Equation (93).

The median PRyMordial response is

$$\bar{\mathbf{c}}_{20}^{\text{PRyMordial}} = (+0.343241, -0.387065, -0.601891)^T. \quad (100)$$

Its direction is stable across the six anchors: the maximum pairwise rotation is approximately  $0.2712^\circ$ . For a direct median-vector comparison, define

$$\delta(\mathbf{a}, \mathbf{b}) = \frac{\|\mathbf{a} - \mathbf{b}\|}{\|\mathbf{b}\|}. \quad (101)$$

Relative to Equation (94), the PRyMordial median gives

| Reference network | Direction cosine | Angular separation | Relative vector difference |
|-------------------|------------------|--------------------|----------------------------|
| PRIMAT            | 0.999999955      | $0.0172^\circ$     | 0.138%                     |
| PArthENoPE        | 0.999981520      | $0.348^\circ$      | 0.832%                     |

Table 10: Direct median-vector comparison of the completed PRyMordial Reaction-20 response with the completed PRIMAT and PArthENoPE authorities.

The three completed networks therefore recover the same  $(+, -, -)$  native Reaction-20 response topology and nearly the same component balance in the shared observable projection. This is a three-network replication of the network-native endpoint response,

not a claim that code-local Jacobians, source kernels, internal normalization coordinates, or complete susceptibility matrices are numerically identical. Across the full PRyMordial 63-reaction response surface, 19 reactions exceed the declared resolved-response norm floor; their median maximum six-anchor direction rotation is  $0.0378^\circ$  and the largest is  $1.0377^\circ$ . Those full-surface stability statistics characterize the numerical authority but are not promoted into a 63-reaction universal-law pass.

## 14.5 Cross-Network Validation Roles

| Network        | Validation role                                                                                        | Publication boundary                                                                                                                                                 |
|----------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRIMAT 0.3.2   | Primary full-reaction authority and completed native source/Jacobian validation                        | Claims restricted to the completed 20,550-row authority surface and the six-anchor first-order mapped light-sector mechanism test                                    |
| PARthENoPE 3.0 | Completed independent full-network Reaction-20 transfer                                                | Claims restricted to the declared shared-endpoint Reaction-20 transfer test                                                                                          |
| PRyMordial     | Third completed 63-reaction native response authority and independent Reaction-20 endpoint replication | 3,030/3,030 rows, 378/378 ladders, and 1,512/1,512 pairs complete; shared-endpoint claims remain restricted to the demonstrated Reaction-20 cross-network comparison |

Table 11: Roles of the three completed validation networks. PRIMAT is the primary full-reaction and native-mechanism authority, PARthENoPE supplies the second independent authority within its declared Reaction-20 transfer test, and PRyMordial supplies a third completed native response authority with independent Reaction-20 endpoint replication.

## 15 Primary PRIMAT Full-Reaction Validation Campaign

The primary numerical campaign used PRIMAT 0.3.2 with its native large network [8]. Every one of the 428 nuclear reactions is perturbed at every baryon-density anchor and every nonbaseline branch. The completed execution count is

$$N_{\text{run}} = 6 + 428 \times 6 \times 8 = 20,550. \quad (102)$$

The six baryon-density anchors are

$$\eta_{10} \in \{6.044, 6.080, 6.094, 6.106, 6.128, 6.134\}, \quad (103)$$

and the branch powers are

$$p \in \{-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2\}. \quad (104)$$

The multiplicative perturbation is

$$f_r(p) = 0.9759799227241401^p, \quad (105)$$

with

$$x = -\ln f_r(p). \quad (106)$$

The resulting ledger contains 10,272 exact matched  $+/-$  branch pairs and 2,568 complete eight-branch reaction–density ladders. Every scientific row retains the complete final abundance vector of 59 tracked nuclides together with nine standard PRIMAT observables. For shared endpoint comparison,

$${}^3\text{He}/H \equiv ({}^3\text{He} + {}^3\text{H})/H, \quad (107)$$

and

$${}^7\text{Li}/H \equiv ({}^7\text{Li} + {}^7\text{Be})/H. \quad (108)$$

The primary production backend is PRIMAT’s C implementation with numerical relative precision  $10^{-10}$  and a 4000-point nuclear-rate grid. The independent Python backend is used for implementation qualification. Scientific rows are process-isolated and committed atomically. At  $\eta_{10} = 6.094$  and  $|p| = 0.5$ , every reaction receives two direct native rate-proof records, one for each sign of the perturbation, giving 856 records covering all 428 reactions and verifying the requested native rate key and perturbation realization.

For every positive finite observable, the preregistered response extraction is

$$\Delta \ln Y = c_1 x + c_2 x^2, \quad (109)$$

with zero intercept and all eight branches used. Heavy or extremely trace species are retained as descriptive measurements when their logarithmic response is numerically defined; they are not promoted post hoc to hard theory gates.

| Campaign layer              | Completed size           | Scientific role                                                              |
|-----------------------------|--------------------------|------------------------------------------------------------------------------|
| Baseline set                | 6 rows                   | Native unperturbed abundance vectors at all $\eta_{10}$ anchors              |
| Full reaction surface       | 20,544 rows              | 428 reactions $\times$ six anchors $\times$ eight symmetric branches         |
| Matched pairs               | 10,272                   | Reciprocity, sign, and local-response consistency                            |
| Complete ladders            | 2,568                    | First- and second-order response extraction                                  |
| Direct rate-proof records   | 856 across 428 reactions | Direct verification of the requested native rate key and perturbation factor |
| Mapped light-reaction layer | 14 reactions             | Direct comparison with the fixed CPTG/PArthENoPE source register             |

Table 12: Completed structure of the primary PRIMAT full-reaction validation campaign.

The completed campaign establishes a high-precision native PRIMAT reaction-response authority surface with complete row, pair, ladder, and direct rate-proof provenance. An independent campaign audit verifies 20,550/20,550 unique row contracts, 10,272/10,272 matched branch pairs, 2,568/2,568 complete eight-branch ladders, and direct rate-key evidence for all 428 reactions.

Independent reconstruction finds zero ledger–commit–request–result identity discrepancies, zero row-hash discrepancies, zero nonempty standard-error records, 20,550/20,550 attempt-1 completions, and finite positive 59-nuclide final states on every row. The preproduction

Python/C backend comparison also passes its frozen  $5 \times 10^{-6}$  numerical agreement requirement, with maximum recorded relative difference  $2.595119 \times 10^{-6}$ .

This authority surface is not repurposed as a fitted universal source-to-observable matrix. CPTG physical validation is supplied separately by the 84-row native source/Jacobian calculation in Section 14.3, whose prediction phase is completed before endpoint-reference exposure.

## 16 Matrix-Exponential Numerical Rigidity Tests

A separate external test suite evaluates the mathematical and software rigidity of the four-sector operator. For a constant generator  $\mathbf{L}$ , finite evolution is computed as

$$\mathbf{X}(t) = \exp(t\mathbf{L})\mathbf{X}(0). \quad (110)$$

The matrix exponential is evaluated with the SciPy linear-algebra implementation, which uses scaling-and-squaring methods associated with the modern matrix-exponential literature [17, 18, 19].

The completed state-evolution campaign contained 1,000,000 randomized trials. Of these, 900,000 were valid-domain states and 100,000 carried adversarial faults. The injected classes included underflow-scale values, nonfinite values, malformed text or type corruption, state-vector bit perturbations, generator perturbations, and conservation-preserving attacks designed to evade a simple mass-sum check.

The primary baryon firewall is

$$\left| \mathbf{1}^T \mathbf{X}(t) - \mathbf{1}^T \mathbf{X}(0) \right| \leq 2 \times 10^{-11}. \quad (111)$$

The tested valid-state domain and fail-closed tolerances are

$$\mathbf{X}(0), \mathbf{X}(t) \in \mathbb{R}^4, \quad \mathbf{L} \in \mathbb{R}^{4 \times 4}, \quad (112)$$

with every component finite, and

$$X_i(0) \geq 0. \quad (113)$$

$$\left| \mathbf{1}^T \mathbf{X}(0) - 1 \right| \leq 10^{-12}. \quad (114)$$

$$X_i(t) \geq -2 \times 10^{-13}. \quad (115)$$

$$X_i(t) \leq 1 + 2 \times 10^{-11}. \quad (116)$$

All 900,000 valid states completed without an unexpected failure, and all 100,000 injected faults were intercepted. No invalid state was silently accepted. The independent scalar curvature-to-current campaign evaluated 2,000,000 valid-domain points with zero nonfinite results, monotonicity violations, concavity violations, or ordered-pair violations. It also distinguished all 27 nearby scalar-law mutations from the fixed relation.

The tests do not claim that floating-point matrix exponentiation is symbolic exact arithmetic. Their purpose is to avoid cumulative low-order time-stepping drift, measure invariant preservation, and verify that invalid states do not silently pass downstream. A separate high-precision transition-operator comparison found maximum double-precision disagreement of  $2.220446049250313 \times 10^{-16}$  and maximum long-horizon mass drift of  $1.4099832412739488 \times 10^{-14}$ . These results qualify numerical implementation behavior within the tested domain; they do not constitute an independent test of the nuclear theory.

| Test class                 | Evaluated | Passed or intercepted | Unexpected failures |
|----------------------------|-----------|-----------------------|---------------------|
| Valid randomized states    | 900,000   | 900,000               | 0                   |
| Adversarial fault states   | 100,000   | 100,000               | 0                   |
| Scalar valid-domain points | 2,000,000 | 2,000,000             | 0                   |
| Scalar-law mutations       | 27        | 27 detected           | 0                   |

Table 13: Completed numerical-rigidity and fail-closed validation results.

## 17 Preregistration and No-Retuning Principle

The numerical tests follow three publication-level constraints. First, the scalar CPTG law, four-sector coordinates, and exact reaction-source construction are fixed before the corresponding validation responses are used. Second, endpoint reaction responses do not enter the construction of the source geometry or the frozen native-mechanism acceptance thresholds. Third, all 84 PRIMAT variational predictions are completed before the authoritative perturbed-endpoint coefficients are opened, and no post-exposure adjustment of the source geometry, propagation rule, or numerical gate is permitted.

Network-native source-to-observable propagation is treated separately from the universal source geometry. In the completed PRIMAT test, the native source contribution and analytic Jacobian are evaluated without access to the endpoint-reference coefficients, and independent BDF and Radau propagations are compared only after the prediction freeze. The completed PRyMordial campaign is evaluated as an independent native endpoint-response authority: its fixed reaction register, six anchors, eight-branch ladder, and rate-scaling protocol are applied without network-specific geometric refitting, and its raw endpoint vector is compared only in the shared observable space after network evolution. A bare geometric source coordinate is not substituted directly for the network-native endpoint vector. The matrix-exponential stress suite remains a numerical implementation test and is not counted as evidence for nuclear physical correctness.

## 18 Physical Interpretation of the Four-Sector Geometry

The vertex-to-bridge-to-closure-to-saturation sequence is not a post hoc classification imposed on the calculated abundances. It is the physical interpretation of the baryon-normalized state geometry and its exact reaction-incidence structure. Each coordinate represents a distinct level of nuclear organization: unbound nucleons, the first bound pair, the charge-resolved mass-three sector, and the saturated helium-four endpoint. The ordered sequence therefore describes the progressive transfer of baryon content into increasingly complete bound structures while preserving the internal charge information required to distinguish physically different reaction directions.

This ordering does not require every reaction trajectory to proceed irreversibly from left to right. Forward and reverse reactions remain part of the network. Rather, the sequence defines the structural orientation of the state space in which those reactions act. The incidence operator measures transport between successive structural levels, while the polarization coordinate retains information that cannot be represented by baryon content alone.

### 18.1 Vertex

The free-neutron and free-proton populations form the unbonded source sector of the light-element network. Their combined baryon coordinate is

$$X_V = Y_n + Y_p^{\text{free}}, \quad (117)$$

while their internal charge orientation is retained by

$$P_V = Y_p^{\text{free}} - Y_n. \quad (118)$$

This separation is essential because two free-nucleon states can contain the same total baryon abundance while possessing different neutron-to-proton compositions and therefore different reaction propensities. The vertex is consequently more than an initial abundance reservoir: it is the unresolved source from which every bound light-element channel originates.

### 18.2 Bridge

Deuterium is the first bound state in the sequence and the minimal nuclear connection between the free-nucleon vertex and the higher bound sectors. It contains one neutron and one proton, so its formation converts unbound baryon content into the first composite channel without introducing an internal charge-polarization degree of freedom.

Geometrically, deuterium occupies the bridge coordinate because it carries transport away from the free-nucleon sector while remaining directly connected to both dissociation back into the vertex and capture into the mass-three sector. Its abundance is therefore sensitive

to source-current amplitude, reaction timing, and the competition between forward and reverse transport. The bridge is not merely an intermediate isotope; it is the unique two-body link through which the four-sector light-element path becomes connected.

### 18.3 Closure

The mass-three sector is formed by the mirror pair

$$(T, {}^3\text{He}). \quad (119)$$

Its coarse baryon coordinate,

$$X_C = 3(Y_T + Y_{{}^3\text{He}}), \quad (120)$$

measures the total baryon content stored in mass-three nuclei, while

$$P_C = 3(Y_{{}^3\text{He}} - Y_T) \quad (121)$$

retains the internal charge orientation of that sector.

This is the closure coordinate because the first bound bridge has developed into a three-nucleon structure with an independently resolved polarization direction. The mass-three pair closes the minimal bridge into a charge-sensitive nuclear sector while preserving distinct pathways toward dissociation, exchange, and helium-four production. The separation of  $X_C$  from  $P_C$  allows the universal geometry to represent changes in total mass-three content independently of changes in its tritium–helium-three composition.

### 18.4 Saturation

Helium-four is the saturated terminal state of the four-sector light-element backbone. Its baryon coordinate is

$$X_S = 4Y_{{}^4\text{He}}, \quad (122)$$

and it appears as the final output of the ordered incidence operator. Within the bonded sector, helium-four contains nearly all of the baryon-normalized abundance, demonstrating that the sequence terminates in a strongly concentrated structural endpoint.

Saturation does not mean that every nuclear reaction ceases at helium-four or that transport into heavier nuclei is forbidden. It means that, within the hydrogen–deuterium–mass-three–helium-four state space, helium-four is the terminal concentration of bound baryon content. Its near-unity share of the bonded-sector coordinate follows from the network response and the state geometry; it is not imposed as a fitted abundance condition.

Taken together, the four coordinates describe a single geometric progression from an unbonded source, through the first bound connection, into a charge-resolved intermediate sector, and finally into a saturated light-nuclear endpoint. This physical hierarchy explains why the four sectors can be governed by one transport-polarization architecture without assigning an independent geometric law to each reaction channel.

## 19 Applications Within the Validated Domain

Complete reaction networks remain the reference calculation for nonlinear abundance evolution [20, 8, 9, 10]. The CPTG construction supplies a compact translation layer only after the fixed scalar law and reaction-source coordinates are specified and the network-native observable pushforward has been validated for the requested domain. Within that domain, repeated local perturbative queries can be evaluated through direct scalar and matrix-vector operations rather than by rerunning a complete native network for every nearby input.

The most immediate applications are reaction-rate sensitivity studies, uncertainty propagation, cross-code consistency checks, and experimental-design searches. No performance, plasma-control, or facility-level claim is made in this paper; those applications require separate physical models, hardware benchmarks, and domain-specific validation.

## 20 Downstream Mass-Sector Extension and Reachability

The exact four-sector theorem and its native response validation remain light-sector results. A separate computational companion has nevertheless tested whether the validated program can be carried downstream into a much larger mass-sector construction once a native PRIMAT trajectory is fixed [5]. In the frozen PRIMAT large-network configuration used by that companion, the native inventory contains 59 nuclear species and reaches a maximum mass number  $A = 23$ . A preregistered rule then constructs exactly fourteen native-to-external radiative-capture source edges. Every external state begins at exact zero, and the external graph is rebuilt from a fresh reaction library rather than from a pre-existing heavy-sector endpoint vector. No artificial heavy seed or abundance floor is introduced.

Within the selected forward capture and beta-minus operator, the external transport was independently qualified with a positivity-preserving exponential-inflow method and a triangular backward-Euler verification method on fresh  $64 \rightarrow 128 \rightarrow 256$  sequences at eight preregistered checkpoint masses. The aggregate observed orders are 1.989305 and 0.975679, respectively. The independently extrapolated checkpoint continua agree to a maximum symmetric relative difference of 0.206809%, and the 256-step primary checkpoint values lie within 0.064104% of their Richardson limits. The final six-anchor campaign separately passes finite/nonnegative-state, exact-zero-source, full-vector source-linearity, graph-identity, and reconstruction controls, while every mass-sector sum from  $A = 24$  through  $A = 119$  is structurally reachable and strictly positive at all six PRIMAT anchors.

These results strengthen the operational meaning of scalability used in this article. The same discipline of fixed source construction, preregistration, independent numerical realization, positivity, and fail-closed validation can be propagated far beyond the four light sectors without selecting a desired heavy endpoint. The result is therefore evidence for downstream reachability and numerical transport scalability of the CPTG-anchored architecture, not a proof that every heavy-nucleus response lies in the exact light-sector rank-four susceptibility space.

The authority boundary remains explicit. PRIMAT supplies the native trajectory and endpoint authority only through  $A = 23$  in that frozen campaign;  $A = 24$ –119 are prescribed-PRIMAT-bath reduced-graph transport populations. They are not native PRIMAT heavy-network evolution, are not the endpoint of a self-consistent coupled primordial heavy-element network, and are not asserted to be precision or observationally significant primordial heavy-element yields.

A preregistered  $1024 \rightarrow 2048$  finer-frontier calculation passes at all six PRIMAT anchors through the natural reachable frontier at  $A = 338$ . The global primary  $1024 \rightarrow 2048$  relative change is 0.001520890866, the backward-Euler verification change is 0.030301545857, the independent continuum cross-method difference is 0.001720763239, and the  $P2048$ -to-Richardson difference is 0.000510366163. The primary observed order remains approximately second order,  $p \simeq 1.9944$ , while backward Euler remains approximately first order,  $p \simeq 0.975$ . The  $A = 338$  tail is positive at every anchor, and the selected graph classifies  $A = 339$  as a disconnected rate island rather than a numerically unresolved continuation.

This post- $A = 119$  result is a reduced-graph diagnostic. The preregistered source-tail campaign passes all 18/18 retained-source cases, including all 12/12 hard-gated cases, and the maximum unrenormalized  $A = 338$  truncated-to-baseline departure is  $2.3592239273284576 \times 10^{-13}$ . Neither result is used to enlarge the native PRIMAT heavy-network claim, the exact four-sector theorem, or any precision primordial heavy-element-yield statement.

## 21 Limitations

### Validated Domain

The fixed scalar law and validated exact four-sector geometric construction apply to the stated free-nucleon, deuterium, mass-three, and helium-four sectors and to the reaction, baryon-density, and perturbation domains explicitly tested. The exact rank-four source space is a statement about the defined transport-polarization coordinate; it is not by itself a proof that every heavy-nucleus endpoint response has four numerical degrees of freedom.

The logarithmic generator is defined only inside the domain in Equation (50). The mass-seven survival rule is prospectively validated as a separate live source-network abundance-sector extension rather than an additional simplex coordinate. The completed  $A = 1$ –119 companion is used here only as downstream reachability and numerical-scalability evidence; it is not used to infer a heavy-sector rank-four susceptibility or to alter the native light-sector validation.

### Cross-Code Shared-Endpoint Boundary

Cross-network comparison is restricted to observables and source directions that have a common physical definition. PRIMAT supplies the primary completed full-reaction authority and the completed six-anchor native source/Jacobian validation in Table 8.

PArthENoPE establishes the second independent six-anchor Reaction-20 shared-endpoint transfer in Table 7. PRyMordial supplies the third completed native response authority and reproduces the same Reaction-20 endpoint direction in Tables 9 and 10. The three-network result concerns the network-native observable response after dynamical pushforward; it does not identify the bare source coordinate with the endpoint susceptibility.

Internal current amplitudes, integration measures, source kernels, Jacobians, network-native pushforward matrices, normalization coordinates, and solver trajectories remain implementation-specific. A successful shared-endpoint prediction does not imply equality of those internal quantities.

## Direct Reaction Data

The commissioned reaction is strongly consistent with the 91 keV pivot, the Moscoso thermonuclear-rate grid, and the covariance-aware Tišma and Turkat comparisons [15, 12, 13]. The exact Mossa/LUNA replay in Tables 4 and 5 places all 13 published  $S$ -factor points from 32.4 to 262.9 keV within the combined  $1\sigma$  uncertainty and all 17 tabulated thermonuclear rates from 0.001 to 10 GK inside their published  $1\sigma$  intervals [16].

In the three-point Dhibar comparison, the measurements at 66.66 and 116.66 keV are consistent, while the 166.66 keV point remains discordant at approximately  $6.74\sigma$  [14]. That isolated point is retained in the reported covariance result, and no theory parameter is changed to accommodate it.

## Network-Native Pushforward and Cross-Network Replication

The universal source-factorization claim does not assert one numerical source-to-observable matrix for all codes. PRIMAT’s completed source-level validation instead propagates the isolated native reaction source through the native analytic Jacobian and compares the resulting endpoint derivative with the independently measured full-network response. The 20,550-row authority surface supplies the endpoint reference measurements; it is not repurposed as a source-kernel fit.

The full-native PRyMordial campaign is complete and closes the third network-authority lane. Its 3,030 committed rows cover all 63 native reactions at the six anchors and eight nonbaseline branches, and its Reaction-20 median endpoint vector agrees with PRIMAT at cosine 0.999999955 and with PArthENoPE at 0.999981520. The cross-network result is three-network replication of the shared Reaction-20 native endpoint response without network-specific geometric refitting. Numerical-rigidity tests establish invariant preservation and fail-closed software behavior; they do not substitute for physical network validation.

## 22 Data, Code, and Evidence Availability

External scientific sources and the PRIMAT, PArthENoPE, and PRyMordial software authorities are cited at their literature or software sources throughout this article. CPTG-generated calculations are associated with named evidence archives, result objects, and checksum identities where an immutable archival release has been established. Distribution locations may serve as mirrors; for cryptographically sealed evidence lanes, the archive filename together with its SHA-256 digest defines the immutable evidence identity.

### 22.1 PRIMAT Full-Coordinate Primordial-Network Evidence

The PRIMAT baryon, radiation, and mass-seven full-coordinate validation is bound by the following authority lock and recognized final-evidence archive.

#### Authority lock

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip`

#### SHA-256

`43b65036e229dc5f04d445f5890399236aaf546c598f1711e572043db6b05bec`

#### Recognized final-evidence archive

`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_20260819_2344EDT_r05_FINAL_EVIDENCE.zip`

#### SHA-256

`9c44543cd2497e72170c0a64173423c9b3bf2f1f3924c9d4e77c16a318fb8b4c`

The authority lock binds the full-coordinate evidence lane to the recognized final-evidence archive and preserves the preregistered baryon/radiation mapping, native-network execution, mass-seven transport result, all 23 scientific gates, and the associated independent audit record.

### 22.2 PRIMAT First-Order Reaction-Response Evidence

The PRIMAT first-order network-conditioned reaction-response validation is preserved separately from the full-coordinate mass-seven evidence.

#### Primary validation archive

`CPTG_PRIMAT_PaperReadyValidationEvidence_20260817_r02.zip`

#### SHA-256

`b41c8ee49477d8330deb56c85a230d8dcc1d4bba80a1405251a4d6ea7f9b3205`

## **Frozen prediction object SHA-256**

679e3e6fc7432c0592004b9bf0985b0756323ceb327050f3303ca6c816f31bff

## **Accepted six-anchor source/Jacobian results**

CPTG\_PRIMAT\_SIXANCHOR\_NATIVE\_SOURCE\_JACOBIAN\_RESULTS\_20260817-202839.zip

## **SHA-256**

8511f3f79d4298d077a32625859440e01f3b530a3eda6a3e999582e1f938baf8

These identities bind the frozen source/Jacobian predictions to the accepted six-anchor native endpoint-response evaluation and preserve the separation between prediction construction and authoritative endpoint-reference exposure.

## **22.3 PArthENoPE Second-Authority Transfer Evidence**

The independent PArthENoPE Reaction-20 transfer used as the second numerical authority within its declared shared-endpoint scope is preserved in the following authority bundle.

### **PArthENoPE authority bundle**

CPTG\_v129\_r109\_PArthENoPE\_CPTG\_Reaction20\_ReplicationAuthorityBundle\_20260731\_r02.zip

## **SHA-256**

3d42d1cb1d710248841db8d7b1ceafcc7569f092b9b35c9f4114ce9615074cfc

This archive preserves the independent PArthENoPE network-transfer evidence. Its claim remains limited to the Reaction-20 transfer scope defined in the text.

## **22.4 PRyMordial Third-Authority Native-Response Evidence**

The completed PRyMordial 63-reaction campaign provides the third native-response authority lane and the independent Reaction-20 endpoint replication reported in this article.

### **Completed PRyMordial authority-results archive**

CPTG\_PryMordial\_Full63Reaction\_CandidateCRE\_RateProofV2\_ControlPlaneRootRepair\_20260823\_r21\_RESULTS\_CURRENT.zip

## **SHA-256**

6f40bb09d11791ac4bd9c89d5be9816d70c70399430b26d0f3a8d2f7fad856f5

This archive preserves the completed 3,030-row native response surface, the reaction-rate proof records, and the Reaction-20 cross-network endpoint result within the claim boundaries stated in the validation sections.

## 22.5 Downstream Reduced-Graph Reachability and Source-Tail Evidence

The downstream PRIMAT-anchored reduced-graph calculations are preserved independently from the native light-sector authority. They support the stated reachability, frontier-qualification, and source-tail robustness claims without extending the exact light-sector rank-four theorem to a native heavy-element network.

### **$A = 1$ –119 reduced-graph evidence**

CPTG\_PRIMAT\_A1A119\_FINAL\_EVIDENCE\_20260818-071825\_AUDITED\_SELFCONTAINED\_r01.zip

### **SHA-256**

39549f1eba0201999aa0953d9ed8b36d4526c5df3ef70c0f11cdc9ae57449620

### **Post- $A = 119$ finer-frontier qualification archive**

CPTG\_PRIMAT\_PostA119\_A160A338\_FinerFrontierContinuumQualification\_FINAL\_EVIDENCE\_20260818\_r01.zip

### **SHA-256**

b0733da182a4a31a810ccd97e23b33389e11099b5b8bfc72e32501157e7dd68c

### **Post- $A = 119$ source-tail robustness archive**

CPTG\_PRIMAT\_PostA119\_A160A338\_SourceTail\_FINAL\_SELFSEALED\_EVIDENCE\_20260819\_r01.zip

### **SHA-256**

d78aae075d3fe579874d3002fdee6b4d47d98712df9dd1ff85d2f9022e4f246a

The source-tail archive verifies 18/18 retained-source rows and 12/12 hard-gated rows, with a maximum unrenormalized  $A = 338$  departure of  $2.3592239273284576 \times 10^{-13}$ . None of these downstream evidence lanes is used to construct the four-sector formulas.

## 22.6 Evidence Identity and Distribution

For cryptographically sealed evidence lanes, reproducibility is anchored by three elements:

1. the named evidence archive or result object;
2. its SHA-256 digest; and
3. the corresponding calculation, freeze rule, or validation scope stated in this article.

The public CPTG research repository provides the distribution location for the papers, evidence packages, and supporting materials associated with this work [6]. Repository paths are access mechanisms rather than scientific identities; where a checksum-bound archive is listed above, the archive filename and SHA-256 digest provide the immutable reference for independently obtained copies.

## 23 Validation Summary

| Contract                                               | Status                                                             | Key result                                                                                                                                                                                                                    |
|--------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalar CPTG law                                        | Analytic form fixed                                                | Held fixed throughout the completed PRIMAT native-mechanism validation; no standalone scalar-error metric is claimed                                                                                                          |
| Exact state geometry                                   | Complete                                                           | Four-sector baryon simplex, ordered incidence operator, and invertible six-species coordinate                                                                                                                                 |
| PRIMAT light-sector closure                            | Complete for six unperturbed baselines                             | Maximum omitted $A > 4$ baryon fraction $2.8937281459917585 \times 10^{-9}$                                                                                                                                                   |
| Reaction-source basis                                  | Complete                                                           | Three transport coordinates plus one charge-constrained polarization coordinate; exact rank four                                                                                                                              |
| Commissioning-channel empirical projection             | Complete within stated scope                                       | Mossa/LUNA, pivot, rate-grid, Tišma, Turkat, and retained discordant Dhibar comparison                                                                                                                                        |
| PArthENoPE Reaction-20 transfer                        | Complete                                                           | 695/695 rows; direction cosine 0.999979437501036; all declared component and six-anchor stability gates pass                                                                                                                  |
| PRIMAT full-reaction authority                         | Complete                                                           | 20,550/20,550 rows; 428 reactions; six anchors; eight branches; 10,272/10,272 matched pairs; 2,568/2,568 ladders; 856 direct rate-proof records covering all 428 reactions; zero row-identity or hash discrepancies           |
| PRIMAT native source/Jacobian validation               | Complete                                                           | 84/84 predictions frozen before endpoint-reference exposure; 48/48 primary resolved tests pass; worst vector discrepancy 0.0367629%; minimum cosine 0.99999999209394; maximum BDF/Radau discrepancy $5.824823 \times 10^{-7}$ |
| PRIMAT-native mass-seven live transport                | Complete                                                           | Full baryon+radiation coordinate: 23/23 preregistered gates pass; survival deviation from $1/\pi$ is 0.0177852%                                                                                                               |
| PRIMAT-anchored downstream reachability                | Complete through $A = 119$ within selected reduced graph           | 14 boundary crossings; exact-zero external initialization; all $A = 24$ –119 sectors reachable and positive at six anchors; continuum cross-scheme difference 0.206809% at preregistered checkpoints                          |
| Post- $A = 119$ finer-frontier continuum qualification | Complete through reachable $A = 338$ within selected reduced graph | 6/6 anchors pass at $1024 \rightarrow 2048$ ; global primary change 0.152089%; cross-method difference 0.172076%; $P_{2048}$ –Richardson difference 0.051037%; $A = 339$ remains a disconnected rate island                   |
| Post- $A = 119$ source-tail robustness                 | Complete within selected reduced graph                             | 18/18 retained-source cases pass; 12/12 hard-gated cases pass; maximum unrenormalized $A = 338$ departure $2.3592239273284576 \times 10^{-13}$                                                                                |
| PRyMordial 63-reaction native authority                | Complete                                                           | 3,030/3,030 rows; 63 reactions; 378/378 ladders; 1,512/1,512 matched pairs; rate-proof worst error $4.44 \times 10^{-16}$ ; Reaction-20 median cosine 0.9999999955 to PRIMAT and 0.999981520 to PArthENoPE                    |
| Numerical implementation rigidity                      | Complete within stated domain                                      | Randomized valid-state, adversarial-fault, scalar-domain, and mutation-detection tests                                                                                                                                        |

Table 14: Validation status of the analytic, empirical, cross-network, mass-seven, and downstream-scalability components of the nuclear-reaction program.

## 24 Conclusion

The results establish a fixed geometric formulation for the free-nucleon, deuterium, mass-three, and helium-four sectors and provide independent tests of whether that formulation transfers between primordial reaction networks. The four-sector state has an exact baryon-conserving incidence representation, and resolving the free-nucleon and mass-three mirror sectors exposes one charge-constrained polarization degree of freedom. Together with the three ordered transport directions, this gives an exact rank-four reaction-source coordinate within the declared light-element domain.

The resulting architecture is

$$\begin{array}{ccc}
 & \text{fixed CPTG scalar curvature law} & \\
 & \downarrow & \\
 & \text{exact reaction transport-polarization source} & \\
 & \downarrow & (123) \\
 & \text{network-native observable pushforward} & \\
 & \downarrow & \\
 & \text{measured abundance response.} &
 \end{array}$$

PRIMAT provides the primary native numerical authority. Its large-network campaign contains 20,550/20,550 committed rows across all 428 reactions, six baryon-density anchors, and eight nonzero perturbation branches, with complete pair and ladder coverage, direct native rate-key evidence for every reaction, zero row-identity or hash discrepancies, and finite positive 59-nuclide final states on every row.

The PRIMAT native source/Jacobian test validates the physical source-to-endpoint mechanism rather than merely the endpoint authority surface. All 84 reaction–anchor predictions were frozen before endpoint-reference exposure. The 48/48 primary resolved tests pass with worst endpoint-vector discrepancy 0.0367629%, minimum direction cosine 0.99999999209394, and maximum BDF/Radau disagreement  $5.824823 \times 10^{-7}$ ; source-direction and baryon residuals remain at approximately machine precision. This supports the factorization of fixed reaction-source geometry through a network-native dynamical pushforward without fitting the perturbed endpoint coefficients.

The locked mass-seven abundance-transport law has a prospective full-coordinate native validation in the primary network. At the jointly transported CPTG baryon and radiation coordinate, all 23/23 preregistered scientific gates pass; the live  ${}^7\text{Li}/{}^7\text{Be}$  sink gives survival 0.318366498376680, only 0.0177852% from the fixed  $1/\pi$  target, while  $\text{D}/\text{H}$ ,  $Y_p$ ,  ${}^3\text{He}/\text{H}$ , and  $N_{\text{eff}}$  remain stable at the declared control level. The result validates the fixed law with the CPTG radiation coordinate included, without turning it into a fitted lithium correction or a fifth four-sector coordinate.

The PRIMAT-anchored reduced-graph analysis extends the validated workflow down-

stream. Starting from a frozen native authority through  $A = 23$ , fourteen boundary captures feed a fresh selected forward graph initialized at exact zero, whose modeled mass-sector support is gap-free and strictly positive through  $A = 119$  at all six anchors. Its two-scheme asymptotic qualification shows near-second-order primary and near-first-order verification convergence, with independently extrapolated checkpoint continua agreeing within 0.206809%. The finer-frontier calculation passes all six anchors through the natural reachable frontier at  $A = 338$ , with a global  $1024 \rightarrow 2048$  primary change of 0.152089%, cross-method continuum difference of 0.172076%, and  $P2048$ –Richardson difference of 0.051037%;  $A = 339$  remains a disconnected rate island in the selected graph. The preregistered source-tail campaign passes all 18/18 retained-source cases, including all 12/12 hard-gated cases, with maximum unrenormalized  $A = 338$  departure  $2.3592239273284576 \times 10^{-13}$ . The post- $A = 119$  program therefore remains a qualified downstream diagnostic rather than a native heavy-network abundance solution or a proof of heavy-sector rank-four universality.

ParthENoPE is the second independent authority within its declared full-network Reaction-20 transfer, and the completed PRyMordial 63-reaction campaign supplies the third native response authority. PRyMordial reproduces the same Reaction-20 (+, −, −) endpoint topology and closely matches both comparison networks, with median-vector cosine 0.999999955 to PRIMAT and 0.999981520 to ParthENoPE. The combined evidence establishes an exact light-sector rank-four source geometry, a PRIMAT-native first-order source-to-endpoint validation, a PRIMAT-native mass-seven transport validation, and three-network Reaction-20 endpoint replication after network-native dynamical push-forward. Downstream PRIMAT-anchored calculations further demonstrate reachability through  $A = 119$  and a numerically qualified diagnostic frontier through  $A = 338$  with source-tail robustness within the selected reduced graph. None of these results requires equality of code-local numerical susceptibility matrices or extends the exact light-sector theorem to heavy nuclei.

## A Core Formula Register

This register collects the analytical formulas that define the four-sector geometry, the completed PRIMAT native-mechanism validation, and the locked mass-seven abundance-transport extension. Network-native numerical pushforward coefficients are intentionally excluded because no single numerical matrix is assumed to be universal across codes.

**Geometric transport ratio.**

$$G_T = \frac{3 - \pi/100}{\pi}.$$

**Acoustic reference and transported BBN coordinates.**

$$\eta_{10}^{\text{ac}} = 6.170380167710, \quad \eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{G_T} \simeq 5.998071834744.$$

**Locked mass-seven live-channel transport.**

$$Y_7 = Y_{7\text{Li}} + Y_{7\text{Be}},$$

$$\begin{aligned} \left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{CPTG}} &= \left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{native}} - \Gamma_7(t) Y_{7\text{Li}}, \\ \left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{CPTG}} &= \left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{native}} - \Gamma_7(t) Y_{7\text{Be}}, \end{aligned}$$

$$\int \Gamma_7(t) dt = \ln \pi, \quad f_7 = \frac{1}{\pi}.$$

**First- and second-order scalar curvature coefficients.**

$$J_{\text{CPTG}}(\eta_{10}) = \frac{1}{\sqrt{8}} \sqrt{\frac{\eta_{10}}{\eta_{10}^{\text{ac}}}},$$

$$K_{\text{CPTG}}(\eta_{10}) = -\frac{3}{16} \frac{\eta_{10}}{\eta_{10}^{\text{ac}}} G_T^{-1/3}.$$

**Logarithmic perturbation coordinate and prescribed PRIMAT ladder.**

$$x = -\ln f_r, \quad f_r(p) = 0.9759799227241401^p.$$

**Four-sector state and ordered incidence operator.**

$$\mathbf{X} = (X_V, X_B, X_C, X_S)^T, \quad \mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \frac{d\mathbf{X}}{dx} = \mathbf{D}\mathbf{j}.$$

**Charge-constrained polarization and reaction-source vector.**

$$3\delta P_V + \delta P_C = 0, \quad \Pi = \delta P_V = -\frac{\delta P_C}{3},$$

$$\mathbf{s}_r = (q_{VB}, q_{BC}, q_{CS}, \Pi)^T.$$

**Network-native source-kernel expansion and source drives.**

$$\mathcal{F}_{r,\mathcal{N}}(x) = \mathcal{A}_{r,\mathcal{N}}^{(1)}x + \mathcal{B}_{r,\mathcal{N}}^{(2)}x^2 + \mathcal{O}(x^3),$$

$$\mathbf{d}_{r,\mathcal{N}}^{(1)} = \mathcal{A}_{r,\mathcal{N}}^{(1)}\mathbf{s}_r, \quad \mathbf{d}_{r,\mathcal{N}}^{(2)} = \mathcal{B}_{r,\mathcal{N}}^{(2)}\mathbf{s}_r.$$

**Validated first-order network-native observable pushforward.**

$$\mathbf{y}_{r,\mathcal{N}}^{(1)} = \mathbf{L}_{\mathcal{N}}^{(1)}(\eta_{10})\mathbf{d}_{r,\mathcal{N}}^{(1)}.$$

The empirical quadratic endpoint coefficient is retained as a second-order diagnostic; no closed second-order relation  $\mathbf{y}^{(2)} = \mathbf{L}^{(2)}\mathbf{d}^{(2)}$  is claimed.

**PRIMAT endpoint reference extraction and native-mechanism gates.**

$$\Delta \ln Y = c_1 x + c_2 x^2, \quad \|\mathbf{c}_{1,\text{ref}}\| \geq 10^{-3} \quad \text{for a primary resolved test,}$$

$$\frac{\|\mathbf{c}_{1,\text{var}} - \mathbf{c}_{1,\text{ref}}\|}{\|\mathbf{c}_{1,\text{ref}}\|} \leq 0.02, \quad \cos \theta \geq 0.999,$$

$$\epsilon_{\text{BDF/Radau}} \leq 5 \times 10^{-4}, \quad \cos \theta_{\text{BDF/Radau}} \geq 0.999999,$$

$$\epsilon_{\text{source}} \leq 10^{-10}, \quad \epsilon_{\text{baryon}} \leq 10^{-10}, \quad \epsilon_{\text{baseline}} \leq 5 \times 10^{-6}.$$

**Generator-domain condition.**

$$1 + \frac{3J_{\text{CPTG}}(\eta_{10})}{G_T^{1/3}}x > 0.$$

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